Software Test Plan (STP) — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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Version: 5.1 · **Date:** 2026-11-12
Status: Executed against build 2d74453 on 3 Nov 2026: 118 of 119 cases Pass, ST_013 failing on Defect 1 (FR-028). Defect 1 fixed and ST_013 re-run on 11 Nov (Deviations D-15): **119 of 119 cases Pass**

Revision history

Version	Date	Author	Change summary
M1	2026-09-06	Kshitij	118 test cases across 13 use cases (68 UT / 37 IT / 13 ST), written to the course test case template
3.0	2026-10-15	Kshitij	Rebuilt to the Part-2 deliverables
3.1	2026-11-01	Kshitij	IT_038 added for NFR-012: 119 cases
4.0	2026-11-03	Kshitij	Actual Result and Test Result recorded for all 119 cases; Deliverable 3 case corrections
4.1	2026-11-03	Kshitij	Arranged to the course's test plan outline; §5.1 Security Validation added; ST_013 Fail on Defect 1
4.2	2026-11-11	Kshitij	ST_013 Pass after the Defect 1 fix
5.0	2026-11-11	Kshitij	Arranged to the course STP template (§1–§15): test items, features (not) to be tested, entry criteria, schedule, risks, assumptions, metrics and approvals added; test cases moved to Appendix A unchanged
5.1	2026-11-12	Kshitij	200-user load test run for NFR-001 and NFR-003 and moved from §4 to §3; load-test environment (§6.2) and risk (§10) added (Deviations D-18)

1. Introduction

1.1 Purpose

This document defines the test plan for the Online Examination Portal: objectives, scope, strategy, resources, schedule and responsibilities. It also records the result of executing every test case.

Objectives.

- Every functional requirement is verified by at least one passing test case (NFR-007).
- Every test case that can be automated is, and runs on every push and pull request in CI.

- No release with a failing test: a red CI run blocks the merge to `main`.
- Security requirements are verified both by automated tests and by manual checks against the running system (§5.1).

1.2 Scope

Testing covers all 30 functional requirements across the 13 use cases, and the 11 non-functional requirements that can be checked on one machine (§3). Coverage is **8–10 test cases per use case**, **119 cases** in all, at Unit, Integration and System level.

The features not tested, and why, are in §4.

1.3 References

- Software Requirements Specification v4.0 — `docs/SRS.pdf` (requirement IDs, acceptance criteria, RTM).
- Software Architecture and Design Specification v3.0 — `docs/SAD.pdf`.
- Use Case Diagram and Use Case Flow — `docs/Use_Case_Diagram.pdf`, `docs/Use_Case_Flow.pdf`.
- System Validation Report, Unit Test Report, Code and Branch Coverage Reports — `docs/`.
- Deviations register — `docs/Deviations.pdf`.
- The course test case template, `Test Plan Document.docx`, which sets the columns of Appendix A.

1.4 Definitions

Term	Meaning
UT / IT / ST	Unit, integration and system test case (§5)
FR / NFR / UC	Requirement and use case IDs of the SRS
SO	Security objective (SRS §5.1.1)
RTM	Requirements Traceability Matrix (SRS §8)
CI	Continuous integration — GitHub Actions running the suite on every push and pull request
S1 … S4	Defect severity (§5.4)
Fixture	A named, seeded test account or record (§6.3)
Frozen build	Commit `2d74453`, the build every case was executed against

2. Test Items

The build under test is commit `2d74453` of `main` (end of Sprint 2), plus the Defect 1 fix of 11 Nov for the ST_013 re-run. Each Django app is a test item:

- `identity` — registration, sign-in, roles, lockout, enrollment, eligibility, DPDP consent and erasure
- `questionbank` — questions, options, tags, versioning
- `examsetup` — examination details and policy, question selection, publish, schedule

- `attempts` — dashboard, start, timer, paper, palette, autosave, resume, submit and auto-submit
- `evaluation` — scoring, percentage, pass/fail
- `results` — withholding, publish and unpublish, scorecard
- `integrity` — focus-loss events, integrity and audit views
- `repository` — models and the append-only audit trail
- `portal` — settings, role middleware, error pages, traceability matrix, page templates

3. Features to be Tested

Features mapped to SRS requirement IDs:

Feature	Requirements	Use cases
Registration, sign-in and role-based access	FR-001, FR-002, FR-003	UC-01, UC-02
Courses and enrollment	FR-004, FR-005, FR-006	UC-03, UC-04
Question bank	FR-007, FR-008, FR-009	UC-05
Examination authoring and publication	FR-010, FR-011, FR-012	UC-06
Scheduling and the Student dashboard	FR-013, FR-014	UC-07
Starting an attempt, timer and palette	FR-015 … FR-018	UC-08
Autosave, resume and auto-submission	FR-019, FR-020, FR-021	UC-09, UC-10
Automatic evaluation	FR-022, FR-023, FR-024	UC-11
Results and scorecard	FR-025, FR-026, FR-027	UC-12
Integrity events, answer-key protection and audit trail	FR-028, FR-029, FR-030	UC-13
Response time and capacity: 200 attempts, 50,000 questions	NFR-001, NFR-003	—
Evaluation time ≤ 5 s	NFR-002	—
Recovery after restart	NFR-004	—
Browsers and 360 px width	NFR-005	—
Every FR has a test case	NFR-007	—
DPDP consent and erasure	NFR-008	—
WCAG 2.1 AA	NFR-009	—
Configurable exam policy	NFR-010	—
Password hashing, lockout, CSRF/ORM/error pages, secrets	NFR-011 … NFR-014	—

4. Features Not to be Tested

- **NFR-006** — a process requirement, checked on the GitHub history rather than by a test.
- **Django, the browsers and SQLite themselves** — assumed tested by their maintainers.
- **Nginx and HTTPS configuration** — no deployment server exists for the course. The load test runs Gunicorn on a workstation instead (§6.2).

- **Everything the SRS puts out of scope** — proctoring, descriptive answers, payments, mobile apps.

5. Test Approach / Strategy

Tests are written with the feature, not after it. Whoever builds a requirement writes its tests in the same pull request.

Each automated test is named after its case ID (`test_UT_001_…`), so a failing test points straight at the case and the requirement behind it. `portal/test_traceability.py` fails the build if a case in this plan has no test.

Levels. Case IDs follow the template's guidance (*"follow some convention like `UT_01`"*):

Prefix	Test level	What it exercises	How it runs
UT_nnn	**Unit test**	One function, validator or rule in isolation — field validation, a scoring calculation, a state transition	Django test runner
IT_nnn	**Integration test**	Two or more components together — view to service to database, page to API	Django test client
ST_nnn	**System test**	An end-to-end flow through the running system as a real actor would drive it	Django test client in CI; re-run by hand in real browsers for the System Validation Report

Numbering is a single running sequence per prefix, so an ID is never reused if a case is later inserted or retired. Acceptance is the system level re-run in real browsers, recorded in the System Validation Report.

Types.

- **Functional** — every case in Appendix A.
- **Regression** — the whole suite on every push and pull request in CI.
- **Performance** — evaluation time (NFR-002, IT_030); the 200-user Locust load test (NFR-001, NFR-003).
- **Recovery** — server restart during an attempt (NFR-004).
- **Security** — §5.1.
- **Usability and accessibility** — axe audit and a keyboard-only attempt (NFR-009).
- **Portability** — Chrome, Edge and Firefox, and 360 px width (NFR-005).

5.1 Security Validation

How each security requirement is validated, by whom and when. Automated checks run on every push in CI; the manual checks are run against the live system for the System Validation Report.

Security objectives (from `docs/SRS.pdf` §5.1.1):

- **SO-1 Confidentiality of the examination** — no Student sees the answer key, another Student's attempt, or a result before it is published.
- **SO-2 Integrity of attempts and scores** — time limits, scores and the audit trail cannot be altered from the browser.
- **SO-3 Protection of accounts and secrets** — credentials and keys stay safe even if the database or the repository is exposed.

Requirement	Objective	What must be confirmed	Validation method	Test cases / evidence	Owner	Timing	Tools
FR-003 Role-based access	SO-1	Every protected URL refuses other roles (403) and anonymous users (302 / 401)	Automated integration tests; manual URL tampering	UT_011, IT_004, IT_005, IT_006, ST_002	Kshitij	Sprint 1 CI; Deliverable 3 validation	Django test client, Playwright
FR-006, FR-025 Own data only	SO-1	Another course's exam and another Student's result return 403	Automated tests; manual URL tampering	IT_009–IT_011, IT_032, IT_033, ST_004, ST_012	Komal	Sprint 1–2 CI; validation	Django test client, Playwright
FR-029 No answer key in payloads	SO-1	No page or JSON reply of an attempt carries `is_correct` or any key marker	Automated payload test; manual inspection of every payload the browser received	IT_035, ST_013	Kartik	Sprint 1 CI; validation	Django test client, Playwright response capture
FR-017, FR-020 Server-side timing	SO-2	A changed client clock changes nothing; expiry submits at start + duration exactly	Automated tests; manual 3-minute expiry run	UT_042, IT_027, ST_010	Kartik, Kshitij	Sprint 1 CI; validation	Django test client, Playwright
FR-030 Append-only audit	SO-2	Every change writes one entry; no screen or endpoint edits or deletes one	Automated tests; manual reconciliation of the audit log after one exam cycle	UT_068, IT_036, IT_037, ST_013	Komal	Sprint 2 CI; validation	Django test client, Playwright
NFR-011 Password hashing	SO-3	Stored passwords are `pbkdf2_sha256$…` only, never plain text	Automated assertion; manual read of the password column	`identity/test_auth.py`; validation check NFR-011	Kartik	Sprint 1 CI; validation	Django test runner, ORM query
NFR-012 Account lockout	SO-3	The 6th sign-in after 5 failures is refused even with the right password; it succeeds after 15 minutes	Automated test	IT_038 (`identity/test_lockout.py`)	Kshitij (D-09)	Sprint 2 CI	Django test client
NFR-013 CSRF, ORM-only, generic errors	SO-2, SO-3	POST without a token → 403; no raw SQL; a forced error shows no stack trace with DEBUG off	Automated test; static code search; manual forced 404	`portal/test_security.py`; code search for `.raw(`, `cursor(`, `.extra(`; validation check NFR-013	Komal	Sprint 1 CI; validation	Django test client, `git grep`, Playwright
NFR-014 Secrets from the environment	SO-3	No secret in the repository or its history; the app refuses to start without `DJANGO_SECRET_KEY`	Secret scan of the full git history; manual start without the variable	validation check NFR-014	Komal	Sprint 1; validation	`git log -p` scan, `manage.py check`

5.2 Entry criteria

Execution of the plan starts when:

- the build is frozen (commit 2d74453) so the cases cannot drift while they run;

- CI is green on that commit;
- the fixtures of §6.3 can be seeded into a fresh database;
- every case in Appendix A has an automated test, or is marked manual (IT_022).

5.3 Exit criteria

Testing is complete when:

1. every one of the 119 cases has been executed and has a result recorded in this plan;
2. every automated test passes on the frozen build;
3. no S1 or S2 defect is open;
4. statement coverage is at least 80% and branch coverage at least 70% (targets set in 4.1; measured 88.8% and 72.1% on build 2d74453);
5. the System Validation Report records the manual checks of §5.1.

At v5.0 all five criteria are met and no defect is open: Defect 1 (S3) was fixed on 11 Nov and ST_013 passes. NFR-001 and NFR-003 pass the 200-user load test on 12 Nov (System Validation Report §4.1).

5.4 Bug triage

Severity	Meaning	Example	Rule
S1 Critical	Wrong score, lost answers, a security control bypassed	An answer key reaching the browser	Blocks the release; fixed before anything else
S2 Major	A requirement not met, no workaround	Results visible before publication	Fixed in the current sprint
S3 Moderate	A requirement met only partly, or wrong evidence produced	Extra focus-loss events on question navigation (FR-028, Defect 1 of the System Validation Report)	Fixed in the next sprint; recorded in Deviations
S4 Minor	Cosmetic, wording, layout	A label without spacing	Fixed when convenient

A defect is raised as a GitHub issue with its severity and the failing case ID.

6. Test Environment

6.1 Testing tools

Tool	Used for
Django test runner (`manage.py test`, Python `unittest`)	All 118 automated cases
coverage.py 7.16 (`coverage run --branch`)	Code and Branch Coverage reports
GitHub Actions (Ubuntu, Python 3.12)	Runs the suite on every push and pull request
Playwright 1.64 (Python) with Chrome, Edge and Firefox	IT_022 timing and the System Validation walkthrough
axe-core 4.10	WCAG 2.1 AA audit of the attempt flow (NFR-009)

Tool	Used for
Locust 2.46	Load script for NFR-001 / NFR-003 (`loadtest/locustfile.py`)
Gunicorn 27	The application server for the load test, as in SAD deployment

6.2 Hardware, software and data

Automated runs	A fresh SQLite test database created by the Django test runner for each run, on Windows 11 (Python 3.10) and in CI (Ubuntu, Python 3.12)
System validation	The application under `manage.py runserver` with DEBUG off, on a throwaway SQLite file seeded with the fixtures below (OEP_DB_PATH), driven in real Chrome, Edge and Firefox
Load test	Gunicorn, 25 workers, in WSL 2 Ubuntu 22.04 on a 12-thread Windows 11 laptop; Python 3.10, Django 5.2; a fresh copy of a SQLite file seeded by `manage.py seed_bank` (50,000 questions, 200 enrolled students) for every run; Locust on the same machine
Client	Windows 11; viewport 1280×800, and 360×800 for NFR-005
Server clock	The host clock, NTP-synchronised; every timing rule reads the server clock only
Data	Seeded fixture set — see §6.3

6.3 Standing test fixtures

Referenced by name in the *Test data* column so precise values are not repeated in every row.

Fixture	Value
STU_A	`stu.a@pesu.pes.edu` / `Passw0rd!A` / roll `PES1UG24CS001`, role Student, active
STU_B	`stu.b@pesu.pes.edu` / `Passw0rd!B` / roll `PES1UG24CS002`, role Student, active
FAC_A	`fac.a@pes.edu` / `Fac#2026A`, role Faculty, assigned to CRS_X
FAC_B	`fac.b@pes.edu` / `Fac#2026B`, role Faculty, assigned to CRS_Y only
ADM_A	`admin@pes.edu` / `Adm#2026A`, role Administrator
CRS_X	Course UE24CS351A / "Database Management Systems" / Semester 5, active
CRS_Y	Course UE24CS342AA5 / "Advanced Virtual Reality" / Semester 5, active
EXM_1	Examination "DBMS Unit Test 1", CRS_X, 60 min, declared total 50, 25 questions, published
Q_SC1	Single-correct MCQ, 4 options, correct = option B, +2 marks, −0.5 negative

7. Test Schedule

Milestone	Date
Test case design: 118 cases for 13 use cases (M1)	2026-09-06
Test Plan rebuilt for the Part-2 deliverables	2026-10-15
Sprint 1 — tests written with each feature, CI on every push	2026-10-19 to 2026-10-23
Sprint 2 — tests written with each feature, CI on every push	2026-10-26 to 2026-11-01
IT_038 added; every case keyed to an automated test or marked manual	2026-11-01

Milestone	Date
Build frozen at 2d74453	2026-11-01
Execution of all 119 cases; Code and Branch Coverage; Unit Test Report	2026-11-03
System validation in real browsers (System Validation Report v1.0 and v1.1)	2026-11-02 to 2026-11-03
Defect 1 fixed; ST_013 re-run	2026-11-11
200-user load test (NFR-001, NFR-003)	2026-10-28 (PR #87) and 2026-11-12

8. Test Deliverables

Deliverable	Where
Test Plan, with Actual Result and Test Result for all 119 cases (this document)	`docs/Test_Plan.pdf`
Test cases (manual and automated)	Appendix A; automated tests in each app's `test_*.py`
Test data	The fixtures of §6.3
Unit Test Report	`docs/Unit_Test_Report.pdf`
System Validation Report, with screenshots	`docs/System_Validation_Report.pdf`, `docs/validation/`
Code Coverage Report	`docs/Code_Coverage_Report.pdf`
Branch Coverage Report	`docs/Branch_Coverage_Report.pdf`
Requirements Traceability Matrix in code	`portal/traceability.py`, checked by `portal/test_traceability.py`
Test execution logs	GitHub Actions, workflow *Tests*
Defect reports	GitHub issues; Defect 1 in the System Validation Report

9. Roles and Responsibilities

Role	Name	Responsibility
Backend & Data, repository owner	Kshitij G Shettigar	Tests for own FRs; CI workflow; Code and Branch Coverage reports; System Validation Report (reassigned on 3 Nov, `docs/Deviations.pdf` D-12); reviews and merges every PR
Frontend & UX	Kartik Basavaraj Meti	Tests for own FRs; review of own 43 cases for Deliverable 3; browser checks of NFR-005 and NFR-009
QA & Docs	Komal S Sajja	Tests for own FRs; review of own 36 cases; Unit Test Report

Ownership of each FR follows the Sprint Plan (`docs/Sprint_Plan.pdf`). Whoever builds a requirement writes its tests in the same pull request.

10. Risks and Mitigation

Risk	Mitigation
The Django test client runs no JavaScript, so a defect in the page scripts passes CI	System cases re-run in real Chrome, Edge and Firefox for the System Validation Report; this is how Defect 1 was found

Risk	Mitigation
SQLite has one writer: under 200 users, one of four load runs reached p95 3.6 s with 2 lock timeouts	WAL mode, IMMEDIATE transactions and a 20 s busy timeout (PR #87); every run reported, not just the passing ones (Deviations D-18); re-run the load test before any real deployment
A case loses its automated test, or a test is keyed to the wrong case	`portal/test_traceability.py` fails the build
The build changes while the cases are being executed	Build frozen at `2d74453` for the execution
A browser cannot be made to switch tabs under automation	Tab switches simulated with window blur and focus events, stated in the System Validation Report
An assertion that depends on random page content (the CSRF token) fails at random	Such assertions rewritten to check the field, not the whole page (PR #106)

11. Assumptions & Dependencies

- The fixtures of §6.3 can be seeded into a fresh database for every run.
- The server clock is synchronised by NTP; every timing case reads the server clock only.
- Current versions of Chrome, Edge and Firefox are available for the browser runs.
- Two decisions remain outstanding with the course instructor, and each leaves specific expected values provisional:

Open question	Test cases affected	Effect if resolved differently
Negative-marking convention — a flat deduction per wrong answer, or a fraction of the question's marks	UT_054, UT_056, IT_031	The expected values assume a **flat** deduction of 0.5. A fractional convention changes each expected total
Whether a Pass at exactly the pass mark is inclusive	UT_057	The expected outcome assumes **inclusive** — 40.0% against a 40% pass mark is a Pass

12. Suspension & Resumption Criteria

Testing is **suspended** when the build will not start, when `main` fails CI, or when an S1 defect is open, because results after it would not be trusted.

It **resumes** once the fix is merged and the full suite is green again on `main`. The suspended cases are then re-run from the start.

13. Test Case Management & Traceability

13.1 Traceability to the SRS

Every test case description opens with the requirement ID it verifies, so this document feeds the test-case column of the RTM (SRS §8) without a separate mapping table.

Identifiers are the `FR-nnn` IDs of the SRS, which are also the backlog items on the GitHub Project. The same matrix is held in code (`portal/traceability.py`) and checked against the test suite by `portal/test_traceability.py`, so it cannot silently go stale.

13.2 Coverage per use case

Use case	Requirements verified	Unit	Integration	System	Total
UC-01 Register a Student account	FR-001	6	2	1	9
UC-02 Sign in and role-based access	FR-002, FR-003, NFR-012	5	5	1	11
UC-03 Manage courses and assign faculty	FR-004	5	2	1	8
UC-04 Enroll students and gate visibility	FR-005, FR-006	5	3	1	9
UC-05 Author question bank entries	FR-007, FR-008, FR-009	6	3	1	10
UC-06 Assemble and publish an examination	FR-010, FR-011, FR-012	6	2	1	9
UC-07 Schedule and surface on the dashboard	FR-013, FR-014	5	2	1	8
UC-08 Start an attempt and receive the paper	FR-015, FR-016, FR-017, FR-018	6	3	1	10
UC-09 Autosave and resume	FR-019, FR-021	4	4	1	9
UC-10 Automatic submission on expiry	FR-020	4	3	1	8
UC-11 Evaluate automatically	FR-022, FR-023, FR-024	6	3	1	10
UC-12 Publish results and scorecard	FR-025, FR-026, FR-027	5	3	1	9
UC-13 Integrity events and audit trail	FR-028, FR-029, FR-030	5	3	1	9
Total	**30 of 30**	**68**	**38**	**13**	**119**

Every use case carries at least eight test cases, and every one of the 30 functional requirements of `docs/SRS.pdf` is exercised by at least one test case — satisfying NFR-007.

13.3 How the result columns are used

Up to v3.1 these two columns were left blank, because no build existed to run the cases against. From v4.0 they hold the results of one execution of the whole plan against the frozen Deliverable 3 build, commit `2d74453`, on 3 Nov 2026:

- **118 cases are automated.** Each automated test is named after its case ID (`test_UT_001_…`), so the *Actual Result* names the test or tests that ran the case and their outcome. The run was `coverage run manage.py test -v 2`: 227 tests, all passing, none skipped. The same run produced the Code Coverage and Branch Coverage reports.
- **IT_022 is manual**, because it measures timing inside a real browser. It was run in Chromium through Playwright against a local server, and its *Actual Result* records the measured times.

A case is **Pass** only when every test keyed to it passed. Hand walkthroughs of the system test cases belong to the separate System Validation Report.

v4.1: the System Validation Report re-ran all 13 system cases in real browsers. ST_013 failed there on Defect 1 (FR-028: moving between questions records extra focus-loss events), which the automated test cannot see because the Django test client runs no JavaScript. ST_013 is therefore recorded as **Fail**, and the plan stood at **118 of 119 Pass**.

v4.2: Defect 1 was fixed on 11 Nov, after the freeze (Deviations D-15). ST_013 was re-run in Chrome through the same script, and passes: 2 focus-loss events for 2 tab switches, unchanged after answering three questions and submitting. The Django suite still passes (227 tests). The plan stands at **119 of 119 Pass**.

Changes to the cases in v4.0

These cases were corrected in the Deliverable 3 test-case review, so that each one describes what the application now does and what its test checks:

Case	Change	Why
IT_006	Protected pages redirect to sign-in; only API endpoints answer HTTP 401	A browser is sent to sign in rather than shown a bare 401
IT_011	Now direct access to another course's examination, refused with 403	IT_010 already covers listing; FR-006 also forbids the direct URL
UT_052	Submitted is marked by the submission timestamp, not a SUBMITTED state	The attempt model has no separate state field
UT_060	Adds the refused publication while the window is open	FR-026 allows publication only after the window closes (PR #95)
UT_068	The audit entry also carries the old and new answer key	FR-030 audits answer-key changes (PR #96)
IT_036	Five classes are User, Course, Question, Exam, Attempt; adds actor, exam-field and publication checks	Matches the audit coverage added in PR #96
ST_006	Notes that scheduling is covered by ST_007	Publication now comes before scheduling

14. Test Metrics & Reporting

Metric	Value (v5.0)
Test cases planned	119 (68 UT / 38 IT / 13 ST)
Test cases executed	119 of 119 (100%)
Test cases passed / failed	119 / 0 (100% pass) — ST_013 failed on 3 Nov and passed on its 11 Nov re-run
Automated tests in the suite	227, all passing, none skipped
Requirement coverage	30 of 30 FRs; the 13 NFRs of §3 all pass
Statement / branch coverage	88.8% / 72.1% (targets 80% / 70%)
Defects found / fixed / open	1 / 1 / 0 — Defect 1, severity S3 (FR-028)
Defect aging	Defect 1: found 3 Nov, fixed 11 Nov — 8 days

Reports.

- Per push and pull request: the GitHub Actions run of the whole suite.
- Per execution: the Actual Result and Test Result columns of Appendix A.
- Final: the Unit Test, System Validation, Code Coverage and Branch Coverage reports of §8.

15. Approvals

Role	Name	Signature / Date
Course Coordinator		
Team — Backend & Data	Kshitij G Shettigar	
Team — Frontend & UX	Kartik Basavaraj Meti	

Role	Name	Signature / Date
Team — QA & Docs	Komal S Sajja	

Appendix A — Test cases by use case

The 119 test cases, in the columns of the course test case template. Each description opens with the requirement it verifies (§13.1).

UC-01 — Register a Student account

Verifies FR-001.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_001	User Registration	FR-001 — Register with all four mandatory fields valid	Registration page reachable; no account exists for the email or roll number	1: Open /register 2: Enter full name, institutional email, roll number, password 3: Click Register	Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll PES1UG24CS111, password Str0ng#Pass	Account created with role Student; message "Registration successful"	As expected. Automated: test_UT_001_register_with_all_fields_valid (identity.test_auth): ok	**Pass**
UT_002	User Registration	FR-001 — Reject submission with a mandatory field left empty	Registration page reachable	1: Open /register 2: Leave Roll Number empty 3: Complete the other three fields 4: Click Register	Name Arjun Rao, email arjun.rao@pesu.pes.edu, roll "" (empty), password Str0ng#Pass	Registration rejected; no account created; inline error "Roll number is required"	As expected. Automated: test_UT_002_roll_number_is_required (identity.test_auth): ok	**Pass**
UT_003	User Registration	FR-001 — Reject a non-institutional email domain	Registration page reachable; accepted domain configured as pesu.pes.edu	1: Open /register 2: Enter a personal email 3: Complete remaining fields 4: Click Register	Email arjun.rao@gmail.com, other fields valid	Registration rejected; error "Please register with your institutional email address"	As expected. Automated: test_UT_003_non_institutional_email_is_rejected (identity.test_auth): ok	**Pass**
UT_004	User Registration	FR-001 — Reject a malformed email address	Registration page reachable	1: Open /register 2: Enter an email with no @ 3: Complete remaining fields 4: Click Register	Email arjun.rao.pesu.pes.edu, other fields valid	Registration rejected; error "Enter a valid email address"; no account created	As expected. Automated: test_UT_004_invalid_email_is_rejected (identity.test_auth): ok	**Pass**
UT_005	User Registration	FR-001 — Reject a duplicate institutional email	STU_A already registered	1: Open /register 2: Enter STU_A's email with a fresh roll number	Email stu.a@pesu.pes.edu, roll PES1UG24CS999	Registration rejected; error "An account with this email already exists";	As expected. Automated: test_UT_005_duplicate_email_is_rej	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3: Click Register		STU_A unmodified	ected (identity.test_auth): ok	
UT_006	User Registration	FR-001 — Reject a duplicate roll number	STU_A already registered	1: Open `/register` 2: Enter a fresh email with STU_A's roll number 3: Click Register	Email `new.student@pesu.pes.edu`, roll `PES1UG24CS001`	Registration rejected; error "This roll number is already registered"	As expected. Automated: `test_UT_006_duplicate_roll_number_is_rejected` (identity.test_auth): ok	**Pass**
IT_001	User Registration	FR-001 — Registration persists a Student row with a hashed password	Test database reachable and inspectable	1: Register a new Student 2: Query the users table for that email 3: Inspect the stored password column	Email `arjun.rao@pesu.pes.edu`, password `Str0ng#Pass`	Exactly one row exists; role is `STUDENT`; the password column holds a salted hash; the literal `Str0ng#Pass` appears nowhere in the database or logs	As expected. Automated: `test_IT_001_registration_stores_a_hash_not_the_password` (identity.test_auth): ok	**Pass**
IT_002	User Registration	FR-001 — Two concurrent registrations of the same roll number produce one account	Test database reachable; no account for roll `PES1UG24CS222`	1: Fire two registration requests with the same roll number simultaneously 2: Query the users table	Two requests, both roll `PES1UG24CS222`, different emails	Exactly one account is created; the other is rejected as a duplicate; no partial or orphaned row	As expected. Automated: `test_IT_002_duplicate_roll_number_creates_only_one_account` (identity.test_auth): ok	**Pass**
ST_001	User Registration	FR-001 — End-to-end registration through to a usable account	TEST environment running	1: Open the portal as an anonymous visitor 2: Register with valid details 3: Sign in with the new credentials 4: Confirm the Student dashboard loads	Name `Arjun Rao`, email `arjun.rao@pesu.pes.edu`, roll `PES1UG24CS111`, password `Str0ng#Pass`	Registration succeeds and the new account can immediately sign in and reach the Student dashboard	As expected. Automated: `test_ST_001_register_sign_in_and_load_student_dashboard` (identity.test_auth): ok	**Pass**

UC-02 — Sign in and reach a role-appropriate dashboard

Verifies `FR-002`, `FR-003`, and the sign-in lockout of `NFR-012`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_007	Authentication	FR-002 — Sign in with correct credentials	STU_A exists and is active	1: Open `/login` 2: Enter email and	`stu.a@pesu.pes.edu` / `Passw0rd!A`	Sign-in succeeds; a session is established;	As expected. Automated:	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				password 3: Click Sign in		the Student dashboard is displayed	`test_UT_007_correct_credentials_establish_a_session` (identity.test_auth): ok	
UT_008	Authentication	FR-002 — Reject a correct email with a wrong password	STU_A exists and is active	1: Open `/login` 2: Enter the correct email and a wrong password 3: Click Sign in	`stu.a@pesu.pes.edu` / `WrongPass1`	Sign-in refused; no session established; message "Invalid email or password" — it does not reveal that the email exists	As expected. Automated: `test_UT_008_wrong_password_is_refused` (identity.test_auth): ok	**Pass**
UT_009	Authentication	FR-002 — Reject an unknown email without disclosing that it is unknown	No account exists for the email used	1: Open `/login` 2: Enter an unregistered email and any password 3: Click Sign in	`nobody@pesu.pes.edu` / `AnyPass123`	Sign-in refused with the identical message and a comparable response time to UT_008, so accounts cannot be enumerated	As expected. Automated: `test_UT_009_unknown_email_has_the_same_message` (identity.test_auth): ok	**Pass**
UT_010	Authentication	FR-002 — Reject sign-in with an empty password	STU_A exists	1: Open `/login` 2: Enter a valid email and leave the password empty 3: Click Sign in	`stu.a@pesu.pes.edu` / "" (empty)	Sign-in refused; inline error "Password is required"; no authentication attempt is recorded	As expected. Automated: `test_UT_010_empty_password_is_refused` (identity.test_auth): ok	**Pass**
UT_011	Authorization	FR-003 — Every account carries exactly one role	STU_A, FAC_A and ADM_A exist	1: Query the stored record for each of the three accounts 2: Inspect the role field	The three fixture accounts	Each holds exactly one role, drawn from Student, Faculty or Administrator; no account holds none or more than one	As expected. Automated: `test_UT_011_every_account_carries_exactly_one_valid_role` (attempts.test_cohorts): ok	**Pass**
IT_003	Authentication	FR-002 — The session identifier is regenerated on sign-in	STU_A exists; browser developer tools available	1: Load `/login` and record the pre-authentication session cookie 2: Sign in successfully 3: Record the cookie again and compare	`stu.a@pesu.pes.edu` / `Passw0rd!A`	The identifier differs before and after; the pre-authentication identifier is invalidated, so a fixated session cannot be reused	As expected. Automated: `test_IT_003_successful_login_rotates_the_session_key` (identity.test_auth): ok	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT_004	Authorization	FR-003 — A Student is refused a Faculty-only API endpoint called directly	STU_A signed in	1: Sign in as STU_A 2: Using the session token, call POST `/api/questions` directly, bypassing the UI 3: Inspect the response	Valid Student session token; a well-formed question payload	Response is HTTP 403; no question is created; the body carries no Faculty data	As expected. Automated: `test_IT_004_student_is_refused_faculty_screens_and_endpoints` (identity.tests): ok	**Pass**
IT_005	Authorization	FR-003 — A Faculty member is refused an Administrator-only endpoint	FAC_A signed in	1: Sign in as FAC_A 2: Call POST `/api/admin/courses` with a valid payload 3: Inspect the response	Valid Faculty session token; a well-formed course payload	Response is HTTP 403; no course is created	As expected. Automated: `test_IT_005_faculty_is_refused_administrator_screens` (identity.tests): ok	**Pass**
IT_006	Authorization	FR-003 — An unauthenticated client is refused every protected endpoint	Application running; no session held	1: Enumerate the application's endpoints 2: Call each with no session cookie, excluding sign-in and registration 3: Record each status code	The full endpoint list	Every protected page redirects to sign-in (HTTP 302 to `/login/?next=…`); every protected API endpoint returns HTTP 401; a view that declares no roles is refused to everyone, so no endpoint defaults to open	As expected. Automated: `test_IT_006_unauthenticated_client_is_refused` (identity.tests): ok	**Pass**
IT_038	Authentication	NFR-012 — Five failed sign-ins within 15 minutes lock the account for 15 minutes	STU_A exists and is active; no failed sign-ins recorded for it	1: Sign in as STU_A with a wrong password five times 2: Sign in with the correct password 3: Advance the recorded failures by 16 minutes 4: Sign in with the correct password again	`stu.a@pesu.pes.edu` / `WrongPass1` ×5, then `Passw0rd!A`	Step 2 is refused with "Too many failed sign-ins. Try again in 15 minutes." even though the password is right, and no session is established; step 4 succeeds	As expected. Automated: `test_IT_038_lock_lifts_after_15_minutes` (identity.test_lockout): ok `test_IT_038_sixth_attempt_with_right_password_is_refused` (identity.test_lockout): ok	**Pass**
ST_002	Authentication & Authorization	FR-002, FR-003 — Each role lands on its own dashboard and	STU_A, FAC_A, ADM_A exist and are active	1: Sign in as STU_A, note the landing page, attempt	The three fixture accounts	Each role lands on its own dashboard; every cross-role navigation	As expected. Automated: `test_ST_002_each_role_lands_on_i`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		cannot reach another's		/faculty/questions, sign out 2: Repeat as FAC_A, attempting /admin/courses 3: Repeat as ADM_A, confirming the admin dashboard loads		is refused with HTTP 403 and an authorisation message; no other role's data is rendered	ts_own_dashboard (identity.tests): ok	

UC-03 — Manage courses and assign faculty

Verifies FR-004.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_012	Course Management	FR-004 — Create a course with code, title and semester	ADM_A signed in	1: Open /admin/courses 2: Click New course 3: Enter code, title and semester 4: Save	Code UE24CS351A, title Database Management Systems, semester 5	The course is created and listed as Active; message "Course created"	As expected. Automated: test_UT_012_admin_creates_a_course (identity.test_courses): ok	**Pass**
UT_013	Course Management	FR-004 — Reject a course with any mandatory field empty	ADM_A signed in	1: Open the New course form 2: Leave the Title empty 3: Complete code and semester 4: Save	Code UE24CS999A, title "" (empty), semester 5	Creation rejected; inline error "Course title is required"; no course created	As expected. Automated: test_UT_013_empty_course_title_is_rejected (identity.test_courses): ok	**Pass**
UT_014	Course Management	FR-004 — Reject a duplicate code within the same semester	CRS_X exists as UE24CS351A in semester 5	1: Open the New course form 2: Enter the same code and semester with a different title 3: Save	Code UE24CS351A, title DBMS Repeat, semester 5	Creation rejected; error "A course with this code already exists for the selected semester"	As expected. Automated: test_UT_014_duplicate_code_and_semester_is_rejected (identity.test_courses): ok	**Pass**
UT_015	Course Management	FR-004 — Reject assigning a non-Faculty account as Faculty	CRS_X exists; STU_A is a Student	1: Open CRS_X 2: Open the Assign faculty picker 3: Attempt to assign STU_A	STU_A = stu.a@pesu.pes.edu	The assignment is rejected; only Faculty-role accounts are selectable, or the attempt raises "Only Faculty accounts can	As expected. Automated: test_UT_015_student_cannot_be_assigned_as_faculty (identity.test_courses):	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
						be assigned to a course"	ok	
UT_016	Course Management	FR-004 — Reject a duplicate faculty assignment	FAC_A already assigned to CRS_X	1: Open CRS_X 2: Assign FAC_A again 3: Inspect the assigned list	FAC_A on CRS_X	The duplicate is rejected or ignored; FAC_A appears exactly once	As expected. Automated: `test_UT_016_assigning_same_faculty_twice_has_one_assignment` (identity.test_courses): ok	**Pass**
IT_007	Course Management	FR-004 — Archiving preserves history while blocking new work	CRS_X has enrolled Students and one completed examination	1: As ADM_A, archive CRS_X 2: Attempt to create a new examination under it 3: Open its past examination's results	CRS_X with EXM_1 completed	New examinations and enrollments are refused with "This course is archived"; the existing examination, attempts and results remain readable	As expected. Automated: `test_IT_007_archived_course_rejects_enrollment` (identity.test_courses): ok	**Pass**
IT_008	Course Management	FR-004 — Assigning a second faculty grants both authoring rights	CRS_X assigned to FAC_A only	1: As ADM_A, assign FAC_B to CRS_X 2: Sign in as FAC_B 3: Create a question under CRS_X	FAC_B newly assigned to CRS_X	CRS_X appears in FAC_B's course list and the question is created successfully	As expected. Automated: `test_IT_008_newly_assigned_faculty_sees_course` (identity.test_courses): ok	**Pass**
ST_003	Course Management	FR-004 — End-to-end course setup by an Administrator	TEST environment running	1: Sign in as ADM_A 2: Create a course 3: Assign FAC_A to it 4: Sign in as FAC_A and confirm the course is listed 5: Archive the course as ADM_A 6: Confirm FAC_A can no longer add examinations to it	Code UE24CS360A, title Operating Systems, semester 5; FAC_A	Every stage behaves as specified; the archived course remains readable but closed to new work	As expected. Automated: `test_ST_003_admin_creates_and_archives_course` (identity.test_courses): ok	**Pass**

UC-04 — Enroll students and gate examination visibility

Verifies FR-005, FR-006.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_017	Enrollment	FR-005 — Faculty enrolls a Student into their own course	FAC_A signed in and assigned to CRS_X; STU_A exists	1: Open CRS_X 2: Click Enroll student 3: Search for STU_A and confirm	STU_A = roll PES1UG24CS001	STU_A appears on the CRS_X roster; message "Student enrolled"	As expected. Automated: `test_UT_017_faculty_can_enroll_a_student` (identity.test_courses): ok	**Pass**
UT_018	Enrollment	FR-005 — Reject a duplicate enrollment	STU_A already enrolled in CRS_X	1: Open CRS_X 2: Enroll STU_A again 3: Inspect the roster	STU_A on CRS_X	The duplicate is rejected with "This student is already enrolled"; the roster shows STU_A once	As expected. Automated: `test_UT_018_duplicate_enrollment_is_rejected` (identity.test_courses): ok	**Pass**
UT_019	Enrollment	FR-005 — Reject enrolling a Faculty or Administrator account	FAC_A signed in on CRS_X	1: Open the Enroll student picker 2: Search for FAC_B and attempt to enroll	FAC_B = `fac.b@pes.edu`	The enrollment is rejected; only Student-role accounts are selectable, or the attempt raises "Only Student accounts can be enrolled"	As expected. Automated: `test_UT_019_faculty_account_cannot_be_enrolled_as_student` (identity.test_courses): ok	**Pass**
UT_020	Enrollment	FR-005 — Faculty cannot enroll into a course they are not assigned to	FAC_B assigned to CRS_Y only	1: Sign in as FAC_B 2: Call the enroll endpoint for CRS_X directly	FAC_B, target CRS_X, student STU_A	The call is refused with HTTP 403; no enrollment is created	As expected. Automated: `test_UT_020_unassigned_faculty_cannot_enroll_students` (identity.test_courses): ok	**Pass**
UT_021	Enrollment	FR-006 — A Student sees only the examinations of enrolled courses	STU_A enrolled in CRS_X only; both CRS_X and CRS_Y have published examinations	1: Sign in as STU_A 2: Open the dashboard 3: List the examinations shown	STU_A; CRS_X and CRS_Y each with one published exam	Only the CRS_X examination is listed; the CRS_Y examination does not appear	As expected. Automated: `test_UT_021_student_sees_only_exams_of_enrolled_courses` (attempts.test_cohorts): ok	**Pass**
IT_009	Enrollment	FR-006 — Direct URL access to a non-enrolled examination is refused	STU_A not enrolled in CRS_Y; a CRS_Y examination exists with a known ID	1: Sign in as STU_A 2: Request the CRS_Y examination URL directly 3: Inspect the response	STU_A; CRS_Y examination ID	Response is HTTP 403; no title, instructions or questions appear in the body	As expected. Automated: `test_IT_009_admin_can_enroll_a_student` (identity.test_courses): ok	**Pass**
IT_010	Enrollment	FR-006 — The start-attempt endpoint refuses a non-enrolled Student	STU_A not enrolled in CRS_Y; a live CRS_Y examination exists	1: Sign in as STU_A 2: Call POST `/api/attempts` for the CRS_Y examination 3: Query the attempts	STU_A; live CRS_Y examination	The call is refused with HTTP 403; no attempt record is created	As expected. Automated: `test_IT_010_draft_exam_is_hidden_from_list` (attempts.test_visibility): ok	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				table			`test_IT_010_enrolled_student_sees_published_exam` (attempts.test_visibility): ok `test_IT_010_unenrolled_student_does_not_see_exam` (attempts.test_visibility): ok `test_IT_010_student_not_enrolled_is_refused` (attempts.tests): ok	
IT_011	Enrollment	FR-006 — An unenrolled Student cannot open another course's examination detail	STU_A not enrolled in CRS_Y; a published CRS_Y examination exists	1: Sign in as STU_A 2: Open the CRS_Y examination's detail URL directly	STU_A; published CRS_Y examination	The request is refused with HTTP 403 Forbidden; no examination detail is shown	As expected. Automated: `test_IT_011_unenrolled_student_gets_403_for_exam_detail` (attempts.test_visibility): ok	**Pass**
ST_004	Enrollment	FR-005, FR-006 — End-to-end cohort isolation between two courses	TEST environment; CRS_X and CRS_Y each with a published examination	1: As FAC_A, enroll STU_A into CRS_X 2: As FAC_B, enroll STU_B into CRS_Y 3: Sign in as STU_A and list visible examinations 4: Sign in as STU_B and do the same 5: From each account, request the other's examination URL	STU_A → CRS_X; STU_B → CRS_Y	Each Student sees only their own course's examination; each cross-course URL request is refused with HTTP 403	As expected. Automated: `test_ST_004_end_to_end_cohort_isolation_between_two_courses` (attempts.test_cohorts): ok	**Pass**

UC-05 — Author and manage question bank entries

Verifies `FR-007`, `FR-008`, `FR-009`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_022	Question Bank	FR-007 — Create a single-correct MCQ with four options	FAC_A signed in and assigned to CRS_X	1: Open `/faculty/questions/new` 2: Enter the stem	Stem `Which normal form removes transitive dependency?`,	The question is saved to the CRS_X bank and appears in the question list	As expected. Automated: `test_UT_022_valid_single_correct_que`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3: Add four options and mark one correct 4: Complete the metadata 5: Save	options 1NF/2NF/3NF/BCNF, correct 3NF		stion_is_saved_and_listed (questionbank.tests): ok	
UT_023	Question Bank	FR-007 — Reject a question with fewer than two options	FAC_A signed in	1: Open the new question form 2: Enter a stem and a single option 3: Save	Stem valid, one option 3NF	Creation rejected; error "A question must have between 2 and 6 options"	As expected. Automated: test_UT_023_one_option_is_rejected (questionbank.tests): ok	**Pass**
UT_024	Question Bank	FR-007 — Reject a question with more than six options	FAC_A signed in	1: Open the new question form 2: Enter a stem and seven options 3: Save	Stem valid, seven options A–G	Creation rejected with the same boundary message; no question is created	As expected. Automated: test_UT_024_seven_options_are_rejected (questionbank.tests): ok	**Pass**
UT_025	Question Bank	FR-007 — Reject a question with no option marked correct	FAC_A signed in	1: Enter a stem and four options 2: Mark none as correct 3: Save	Four options, zero marked correct	Creation rejected; error "Exactly one option must be marked correct"	As expected. Automated: test_UT_025_no_correct_option_is_rejected (questionbank.tests): ok	**Pass**
UT_026	Question Bank	FR-007 — Reject a single-correct question with two options marked correct	FAC_A signed in	1: Enter a stem and four options 2: Mark two as correct 3: Save	Four options, 2NF and 3NF both marked correct	Creation rejected with the same message; exactly one correct option is required	As expected. Automated: test_UT_026_two_correct_options_are_rejected (questionbank.tests): ok	**Pass**
UT_027	Question Bank	FR-008 — Reject a question missing any mandatory metadata field	FAC_A signed in	1: Enter a valid stem and options 2: Leave Difficulty unset 3: Save; repeat for course, topic, marks and negative marks	Stem and options valid; one metadata field empty each time	Each is rejected with an inline error naming the missing field; no question is created	As expected. Automated: test_UT_027_missing_metadata_is_rejected (questionbank.tests): ok	**Pass**
IT_012	Question Bank	FR-008 — Reject a difficulty value outside the permitted set	FAC_A signed in; API reachable	1: Submit a question via the API with an invalid difficulty 2: Inspect the response	Difficulty Impossible	Rejected with HTTP 400 and "Difficulty must be one of Easy, Medium or Hard"	As expected. Automated: test_IT_012_api_rejects_invalid_difficulty (questionbank.tests): ok	**Pass**
IT_013	Question Bank	FR-009 — A question in a published	Q_SC1 is attached to EXM_1, which is	1: Sign in as FAC_A 2: Open Q_SC1	Q_SC1, new stem text	The in-place edit is refused; a **new version**	As expected. Automated:	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		examination cannot be edited in place	published	3: Change the stem 4: Save 5: Query the question table		is created instead; the version attached to EXM_1 is byte-identical to before	`test_IT_013_published_question_is_versioned_not_edited` (questionbank.test_versioning): ok	
IT_014	Question Bank	FR-009 — A question that a Student has answered cannot be deleted	Q_SC1 attached to EXM_1; STU_A has answered it in a submitted attempt	1: Sign in as FAC_A 2: Attempt to delete Q_SC1 3: Inspect the attempt's stored response	Q_SC1, STU_A's submitted attempt	Deletion refused with "This question has been attempted and cannot be deleted"; the stored response remains resolvable to the original question text	As expected. Automated: `test_IT_014_answered_question_cannot_be_deleted` (questionbank.test_versioning): ok	**Pass**
ST_005	Question Bank	FR-007, FR-008, FR-009 — End-to-end authoring, tagging and version-freeze	TEST environment; FAC_A assigned to CRS_X	1: Sign in as FAC_A 2: Author three tagged questions 3: Attach them to a new examination and publish it 4: Attempt to edit one of the three 5: Confirm a new version is created and the published paper is unchanged	Three MCQs across two topics and two difficulties	Authoring and tagging succeed; once published, the paper's questions are frozen — edits produce versions rather than mutations	As expected. Automated: `test_ST_005_author_tag_publish_then_edits_become_versions` (questionbank.test_versioning): ok	**Pass**

UC-06 — Assemble and publish an examination

Verifies FR-010, FR-011, FR-012.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_028	Exam Authoring	FR-010 — Create an examination with all mandatory attributes	FAC_A signed in and assigned to CRS_X	1: Open `/faculty/exams/new` 2: Enter title, course, duration, declared total marks and instructions 3: Save	Title `DBMS Unit Test 1`, course CRS_X, duration 60, total 50, instructions free text	The examination is created in **Draft** state and appears in the Faculty examination list	As expected. Automated: `test_UT_028_exam_defaults_to_draft` (examsetup.tests): ok	**Pass**
UT_029	Exam Authoring	FR-010 — Reject a non-positive duration	FAC_A signed in	1: Open the new examination form 2: Enter a duration of zero	Duration 0, then -30	Both rejected with "Duration must be a positive number of minutes"; no	As expected. Automated: `test_UT_029_duration_must_be_positive` (examsetup.tests): ok	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3: Save; repeat with a negative value		examination created		
UT_030	Exam Authoring	FR-010 — Per-examination policy values persist and are read back	A Draft examination exists	1: Open its settings 2: Enable negative marking at 0.5, enable question and option shuffling, set pass mark 40% 3: Save and reload the page	Negative `0.5`, shuffle both on, pass mark `40`	All policy values persist exactly as set and are shown unchanged after reload	As expected. Automated: `test_UT_030_exam_policies_are_saved` (examsetup.tests): ok	**Pass**
UT_031	Exam Authoring	FR-010 — Reject a pass mark outside 0–100	A Draft examination exists	1: Open its settings 2: Enter a pass mark of 150 3: Save; repeat with −10	Pass mark `150`, then `-10`	Both rejected with "Pass mark must be between 0 and 100"; the stored value is unchanged	As expected. Automated: `test_UT_031_pass_mark_must_be_between_zero_and_100` (examsetup.tests): ok	**Pass**
UT_032	Exam Authoring	FR-012 — A Draft examination is invisible to Students	A Draft examination exists on CRS_X; STU_A enrolled in CRS_X	1: Sign in as STU_A 2: Open the dashboard 3: Request the Draft examination's URL directly	Draft examination ID	The examination does not appear on the dashboard, and the direct request is refused with HTTP 403 or 404	As expected. Automated: `test_UT_032_draft_exam_is_invisible_to_students` (examsetup.test_publication): ok	**Pass**
UT_033	Exam Authoring	FR-012 — Publishing moves the examination out of Draft state	A Draft examination with questions attached	1: Open the Draft examination 2: Click Publish 3: Inspect the stored state	Draft examination with 25 questions	The state changes from DRAFT to PUBLISHED; the transition is recorded with a server timestamp	As expected. Automated: `test_UT_033_publishing_moves_the_exam_out_of_draft_with_a_server_timestamp` (examsetup.test_publication): ok	**Pass**
IT_015	Exam Authoring	FR-011 — Questions are attached from the bank and stored against the examination	FAC_A signed in; CRS_X bank holds at least ten questions	1: Open a Draft examination 2: Select five questions from the bank 3: Save 4: Query the examination-question join table	Five known question IDs	Exactly the five selected questions are attached, in the chosen order, with no duplicates and no unselected question present	As expected. Automated: `test_IT_015_exactly_the_selected_questions_attach_in_the_chosen_order` (examsetup.test_publication): ok	**Pass**
IT_016	Exam Authoring	FR-012 — Publishing makes the examination visible to enrolled Students only	A Draft examination on CRS_X; STU_A enrolled, STU_B not	1: As FAC_A, publish the examination 2: Sign in as STU_A and check the dashboard 3: Sign in as STU_B and	Published examination on CRS_X	STU_A sees it listed; STU_B does not, since visibility requires both publication and	As expected. Automated: `test_IT_016_published_exam_is_visible_to_enrolled_students_only`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				check		enrollment	(examsetup.test_publication): ok	
ST_006	Exam Authoring	FR-010, FR-011, FR-012 — End-to-end paper assembly and publication	TEST environment; CRS_X bank populated; FAC_A assigned	1: Sign in as FAC_A 2: Create a Draft with title, duration and policy values 3: Attach 25 questions from the bank 4: Confirm a Student cannot see it 5: Publish 6: Sign in as an enrolled Student and confirm the paper is listed	60-minute paper, 25 questions, negative `0.5`, pass mark `40`	The paper is invisible while Draft, becomes visible on publication, and carries the configured policy values through unchanged. Scheduling follows publication and is covered by ST_007	As expected. Automated: `test_ST_006_paper_assembly_and_publication_keep_the_policy_values` (examsetup.test_publication): ok	**Pass**

UC-07 — Schedule an examination and surface it on the Student dashboard

Verifies `FR-013`, `FR-014`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_034	Scheduling	FR-013 — Schedule a published examination with a valid window	A published examination of 60 minutes' duration exists	1: Open the examination 2: Set window start and end date-times 3: Save	Start `2026-09-10 10:00`, end `2026-09-10 12:00`	The schedule is saved and displayed; message "Examination scheduled"	As expected. Automated: `test_UT_034_published_exam_window_is_saved_and_shown` (examsetup.test_schedule): ok	**Pass**
UT_035	Scheduling	FR-013 — Reject a window whose end precedes its start	A published examination exists	1: Set an end date-time earlier than the start 2: Save	Start `2026-09-10 12:00`, end `2026-09-10 10:00`	Rejected with "The window end must be later than the window start"; no schedule saved	As expected. Automated: `test_UT_035_window_end_must_be_later_than_start` (examsetup.test_schedule): ok	**Pass**
UT_036	Scheduling	FR-014 — The dashboard classifies an examination as Upcoming before its window	STU_A enrolled in CRS_X; a CRS_X examination scheduled to start tomorrow	1: Sign in as STU_A 2: Open the dashboard 3: Locate the examination	Window starting `now + 24 h`	Listed under **Upcoming** with its start date-time; no Start button is offered	As expected. Automated: `test_UT_036_exam_starting_tomorrow_is_upcoming_without_start_button` (identity.test_dashboard): ok	**Pass**
UT_037	Scheduling	FR-014 — The	STU_A enrolled; a	1: Sign in as STU_A	Window now −	Listed under **Live** with	As expected. Automated:	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		dashboard classifies an examination as Live inside its window	CRS_X examination whose window is currently open	2: Open the dashboard	`10 min` to now `+ 50 min`	an active Start button	`test_UT_037_open_window_is_live_with_start_button` (identity.test_dashboard): ok	
UT_038	Scheduling	FR-014 — The dashboard classifies a submitted examination as Completed	STU_A has one submitted attempt on `EXM_1`	1: Sign in as STU_A 2: Open the dashboard	`EXM_1` with a submitted attempt	Listed under **Completed**; no Start button is offered	As expected. Automated: `test_UT_038_submitted_exam_is_completed_without_start_button` (identity.test_dashboard): ok	**Pass**
IT_017	Scheduling	FR-013 — The stored window is timezone-unambiguous and server-authoritative	A published examination; database inspectable	1: Schedule a window from the UI 2: Query the stored start and end values 3: Change the client machine's timezone and reload	Start `2026-09-10 10:00 IST`	The stored values are unambiguous (UTC or with an explicit offset); the displayed window is unchanged by the client timezone switch	As expected. Automated: `test_IT_017_schedule_is_stored_as_aware_utc_time` (examsetup.test_schedule): ok	**Pass**
IT_018	Scheduling	FR-014 — Dashboard classification is driven by server time, not the client clock	STU_A enrolled; a CRS_X examination whose window opens in one hour	1: Sign in as STU_A 2: Advance the client clock by two hours 3: Reload the dashboard	Window opening at `server now + 1 h`; client clock `+2 h`	The examination remains under **Upcoming**; the tampered client clock does not promote it to Live	As expected. Automated: `test_IT_018_classification_uses_server_time_not_client_clock` (identity.test_dashboard): ok	**Pass**
ST_007	Scheduling	FR-013, FR-014 — End-to-end scheduling and dashboard transitions	TEST environment; a published 30-minute examination; STU_A enrolled	1: As FAC_A, set a 60-minute window opening in two minutes 2: As STU_A, confirm it shows Upcoming 3: Wait for the window to open and refresh 4: Sit and submit the paper 5: Refresh the dashboard	Duration 30, window `60 min`	The examination moves Upcoming → Live → Completed on the Student dashboard as server time advances	As expected. Automated: `test_ST_007_exam_moves_upcoming_live_completed_as_server_time_advances` (identity.test_dashboard_flow): ok	**Pass**

UC-08 — Start an attempt and receive the paper

Verifies `FR-015`, `FR-016`, `FR-017`, `FR-018`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_039	Exam Delivery	FR-015 — Starting is refused before the window opens	STU_A enrolled; EXM_1 window opens in ten minutes	1: Sign in as STU_A 2: Attempt to start EXM_1	Server time `window start − 10 min`	Refused with "This examination has not opened yet"; no attempt record is created	As expected. Automated: `test_UT_039_start_is_refused_before_the_window_opens` (attempts.tests): ok	**Pass**
UT_040	Exam Delivery	FR-015 — Starting is refused after the window closes	STU_A enrolled; EXM_1 window closed one minute ago	1: Sign in as STU_A 2: Attempt to start EXM_1	Server time `window end + 1 min`	Refused with "This examination has closed"; no attempt record is created	As expected. Automated: `test_UT_040_start_is_refused_after_the_window_closes` (attempts.tests): ok	**Pass**
UT_041	Exam Delivery	FR-015 — Starting is refused when a submitted attempt already exists	STU_A has one submitted attempt on EXM_1; the window is still open	1: Sign in as STU_A 2: Attempt to start EXM_1 again	STU_A, EXM_1, one submitted attempt	Refused with "You have already submitted this examination"; no second attempt is created	As expected. Automated: `test_UT_041_start_is_refused_when_a_submitted_attempt_exists` (attempts.tests): ok	**Pass**
UT_042	Exam Delivery	FR-017 — The countdown is computed from server time, not the client clock	STU_A has an attempt in progress on EXM_1, 60-minute duration, 10 minutes elapsed	1: Note the displayed remaining time 2: Set the client clock back by 30 minutes 3: Refresh the attempt page 4: Note the remaining time again	Elapsed `10 min`; client clock moved `−30 min`	Remaining time reads approximately 50 minutes both times; the client clock change gains the candidate nothing	As expected. Automated: `test_UT_042_remaining_time_uses_server_time_for_helper_and_api` (attempts.test_timer_payload): ok	**Pass**
UT_043	Exam Delivery	FR-018 — The palette shows the correct state for each question	STU_A has an attempt in progress on a 25-question paper	1: Start the attempt and inspect question 5's palette state 2: Open question 5, answer nothing, move on, inspect again 3: Return and select an option, inspect again 4: Mark it for review and inspect again	Question 5 across four interactions	The state moves **Not Visited** → **Not Answered** → **Answered** → **Marked for Review**, matching each interaction exactly	As expected. Automated: `test_UT_043_palette_moves_through_every_state` (attempts.test_palette): ok	**Pass**
UT_044	Exam Delivery	FR-018 — Exactly one question is displayed at a time	STU_A has an attempt in progress on a 25-question paper	1: Load the attempt screen 2: Count the question stems rendered in the DOM	25-question paper	Exactly one stem is rendered; the other 24 are not present in the delivered markup	As expected. Automated: `test_UT_044_exactly_one_stem_is_rendered` (attempts.test_palette):	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
							ok	
IT_019	Exam Delivery	FR-016 — The attempt record is created and server-stamped before the paper is delivered	STU_A enrolled; EXM_1 window open; network trace available	1: Click Start 2: Capture the request/response sequence 3: Query the attempts table	STU_A (`stu.a@pesu.pes.edu` / `Passw0rd!A`, enrolled in CRS_X); EXM_1 ("DBMS Unit Test 1", 60 min, 25 questions, published, window open now)	The attempt row exists with a server-generated start timestamp **before** the question payload is returned; no question data is sent if the row cannot be created	As expected. Automated: `test_IT_019_attempt_row_exists_before_the_paper_is_delivered` (attempts.tests): ok `test_IT_019_no_paper_without_an_attempt_row` (attempts.tests): ok	**Pass**
IT_020	Exam Delivery	FR-016 — The stored start time is the server's, not the client's	STU_A enrolled; EXM_1 window open	1: Set the client clock forward by one hour 2: Start the attempt 3: Query the stored start timestamp	Client clock +1 h	The stored start time reflects true server time; the tampered client clock is ignored	As expected. Automated: `test_IT_020_stored_start_time_is_the_servers` (attempts.tests): ok	**Pass**
IT_021	Exam Delivery	FR-018 — Palette state survives navigation and reload	STU_A has an attempt with a mix of palette states across ten questions	1: Produce all four states across ten questions 2: Reload the browser 3: Compare every palette state before and after	Ten questions across four states	Every state is identical after reload, because palette state is derived from server-held data rather than browser memory	As expected. Automated: `test_IT_021_states_survive_reload_because_they_come_from_the_server` (attempts.test_palette): ok	**Pass**
ST_008	Exam Delivery	FR-015, FR-016, FR-017, FR-018 — End-to-end paper delivery under a live window	TEST environment; EXM_1 live; STU_A enrolled	1: Sign in as STU_A 2: Confirm starting is refused before the window opens 3: Start once the window is open 4: Confirm the timer begins at the full duration and counts down 5: Navigate across several questions using the palette 6: Submit	60-minute paper, 25 questions	The attempt starts only inside the window, the server-driven timer counts down correctly, one question shows at a time, and the palette reflects every state transition accurately	As expected. Automated: `test_ST_008_paper_delivery_under_a_live_window` (attempts.test_attempt_flow): ok	**Pass**

UC-09 — Autosave and resume after interruption

Verifies FR-019, FR-021.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_045	Attempt Responses	FR-019 — A response is persisted without any explicit save action	STU_A has an attempt in progress; network trace available	1: Select an option on question 4 2: Take no further action 3: Watch the network trace	Question 4, option A	A persistence request is sent automatically within 5 seconds of the selection; no Save button is required or present	As expected. Automated: `test_UT_045_response_is_persisted_with_no_save_button` (attempts.test_autosave): ok	**Pass**
UT_046	Attempt Responses	FR-019 — Changing a response persists the new value, not the old	STU_A has answered question 3 with option B	1: Return to question 3 2: Select option D 3: Wait six seconds 4: Query the stored response	Question 3, B replaced by D	The stored response reads option D; exactly one response is held for the question	As expected. Automated: `test_UT_046_changing_a_response_keeps_one_row_with_the_new_value` (attempts.test_autosave): ok	**Pass**
UT_047	Attempt Responses	FR-019 — Clearing a response persists the cleared state	STU_A has answered question 3	1: Return to question 3 2: Click Clear response 3: Wait six seconds 4: Query the stored response	Question 3, cleared	The stored response is empty and the palette returns the question to **Not Answered**	As expected. Automated: `test_UT_047_clearing_a_response_stores_it_empty` (attempts.test_autosave): ok	**Pass**
UT_048	Attempt Responses	FR-021 — Remaining time on resume is computed from the original start	STU_A started EXM_1 at server time T; 60-minute duration	1: Note the start time 2: Wait 20 minutes 3: Reload the attempt 4: Read the remaining time	Start T, reload at T + 20 min	Remaining time reads approximately 40 minutes — derived from the original start, not reset by the reload	As expected. Automated: `test_UT_048_remaining_time_on_resume_counts_from_the_original_start` (attempts.test_resume): ok	**Pass**
IT_022	Attempt Responses	FR-019 — The autosave reaches the server inside the 5-second budget	STU_A has an attempt in progress; server logs timestamped	1: Select an option and record the client-side timestamp 2: Locate the corresponding server-side write 3: Compute the difference 4: Repeat for ten selections	Ten selections across ten questions	Every response is committed server-side within 5 seconds of selection; the worst observed case is recorded	As expected. Run by hand in Chromium (Playwright) on 3 Nov against build `2d74453`: 6 of 6 selections were committed server-side; worst case 119 ms, mean 117 ms from selection to server response (budget 5 000 ms)	**Pass**
IT_023	Attempt Responses	FR-019 — A browser crash loses no response	STU_A has an attempt in progress with six questions answered	1: Answer six questions 2: Wait six seconds 3: Kill the browser process	Six answered questions	All six responses are restored; nothing is lost, because persistence did	As expected. Automated: `test_IT_023_saved_responses_survive_a_`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		already autosaved		without submitting 4: Reopen, sign in and resume		not depend on submission	`browser_that_never_submits` (attempts.test_autosave): ok	
IT_024	Attempt Responses	FR-021 — Resuming restores responses and does not grant extra time	STU_A started EXM_1 at T; answered four questions by T + 20 min	1: Answer four questions 2: Disconnect the network for five minutes 3: Reconnect and reopen the attempt 4: Inspect the restored answers and remaining time	Start T, interruption T+20 to T+25	All four answers are restored; remaining time reads approximately 35 minutes, so the interruption grants no extra time	As expected. Automated: `test_IT_024_resume_restores_answers_and_grants_no_extra_time` (attempts.test_resume): ok	**Pass**
IT_025	Attempt Responses	FR-021 — Resuming on a different device restores the same attempt state	STU_A has an attempt in progress on profile A with three answers	1: Answer three questions on profile A 2: Sign in as STU_A on profile B 3: Open the attempt	Three answered questions; two profiles	Profile B resumes the same attempt with the three answers present and time still measured from the original start	As expected. Automated: `test_IT_025_another_device_resumes_the_same_attempt` (attempts.test_resume): ok	**Pass**
ST_009	Attempt Responses	FR-019, FR-021 — End-to-end answer, disconnect, resume and submit	TEST environment; EXM_1 live; STU_A enrolled	1: Start EXM_1 2: Answer five questions, changing two and clearing one 3: Disconnect for three minutes 4: Reconnect and resume 5: Verify all responses and remaining time 6: Answer the rest and submit 7: Inspect the stored attempt	25-question paper; five answered, two changed, one cleared	Every selection, change and clear survives the interruption exactly; remaining time reflects the original start; the submitted attempt records precisely the final state of every question	As expected. Automated: `test_ST_009_answer_change_clear_resume_and_submit` (attempts.test_attempt_flow): ok	**Pass**

UC-10 — Automatic submission on timer expiry

Verifies `FR-020`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_049	Auto-Submission	FR-020 — The attempt submits itself when remaining time	STU_A has an attempt in progress with 30 seconds	1: Answer one more question 2: Take no action and	Remaining `30 s`, then `0`	At zero the attempt is submitted automatically without	As expected. Automated: `test_UT_049_the_`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		reaches zero	remaining	let the timer run to zero 3: Observe the screen		any click; the Student sees "Time is up. Your paper has been submitted."	`paper_submits_itself_at_zero` (attempts.test_auto_submit): ok	
UT_050	Auto-Submission	FR-020 — Auto-submission retains every response persisted up to that instant	STU_A has an attempt with eight answered questions and 20 seconds remaining	1: Confirm eight answers 2: Let the timer expire 3: Query the submitted attempt's responses	Eight answered of 25	All eight responses are present; the remaining 17 are recorded as unattempted	As expected. Automated: `test_UT_050_every_response_saved_before_expiry_is_kept` (attempts.test_auto_submit): ok	**Pass**
UT_051	Auto-Submission	FR-020 — No further response is accepted after auto-submission	STU_A's attempt has just auto-submitted	1: Attempt to select an option on the last displayed question 2: Call the response-save endpoint directly with the old attempt ID	Auto-submitted attempt ID	The UI is read-only; the direct API call is refused with "This attempt has already been submitted"; the stored responses are unchanged	As expected. Automated: `test_UT_051_no_response_is_accepted_after_auto_submission` (attempts.test_auto_submit): ok	**Pass**
UT_052	Auto-Submission	FR-020 — The attempt is recorded as submitted with a server timestamp	An attempt that has just auto-submitted	1: Query the attempt record 2: Inspect its submission timestamp	Auto-submitted attempt	The submission timestamp (`submitted_at`) is set, which is what marks the attempt submitted; it is server-generated and equals the attempt start plus the configured duration	As expected. Automated: `test_UT_052_submitted_at_is_the_start_plus_the_duration` (attempts.test_auto_submit): ok	**Pass**
IT_026	Auto-Submission	FR-020 — Expiry is enforced server-side even if the browser is closed	STU_A has an attempt in progress with one minute remaining	1: Answer two questions 2: Close the browser entirely 3: Wait until after the duration elapses 4: Sign in again and open the examination 5: Query the attempt record	Browser closed at T − `60 s`	The attempt is recorded as submitted with the two answers retained; expiry did not depend on the browser being open	As expected. Automated: `test_IT_026_expiry_does_not_need_the_browser_open` (attempts.test_auto_submit): ok	**Pass**
IT_027	Auto-Submission	FR-020 — A client-side clock change	STU_A has an attempt with two	1: Set the client clock back by one hour	Client clock `−1 h`; real elapsed time	The attempt is auto-submitted at the true	As expected. Automated:	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		cannot defer auto-submission	minutes remaining	2: Let real time pass beyond the true expiry 3: Attempt to answer a question	beyond duration	server expiry; further answers are refused	`test_IT_027_a_client_clock_cannot_defer_expiry` (attempts.test_auto_submit): ok	
IT_028	Auto-Submission	FR-020, FR-022 — An auto-submitted attempt enters evaluation like a manual one	Auto-submission has just occurred for STU_A on EXM_1	1: Let the attempt auto-submit 2: Query the evaluation record within 5 seconds	Auto-submitted attempt on a 25-question paper	The attempt is evaluated automatically with a total score, percentage and Pass/Fail outcome, identical in treatment to a manually submitted attempt	As expected. Automated: `test_IT_028_an_auto_submitted_attempt_is_evaluated` (attempts.test_auto_submit): ok	**Pass**
ST_010	Auto-Submission	FR-020 — End-to-end timed expiry on a short paper	TEST environment; a 3-minute examination with five questions; STU_A enrolled	1: Start the examination 2: Answer three of the five questions 3: Let the full three minutes elapse without submitting 4: Observe the screen at zero 5: Sign in as FAC_A and inspect the attempt	3-minute paper, five questions, three answered	The paper submits itself exactly at expiry, the three answers are retained, the two unanswered are recorded as unattempted, and Faculty sees a normally submitted attempt	As expected. Automated: `test_ST_010_three_minute_paper_end_to_end` (attempts.test_auto_submit): ok	**Pass**

UC-11 — Evaluate a submitted attempt automatically

Verifies `FR-022`, `FR-023`, `FR-024`.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_053	Evaluation	FR-023 — Award full positive marks for a correct response	Scoring engine callable in isolation; `Q_SC1` = +2 / −0.5, correct option B	1: Invoke the evaluator with a response of option B 2: Read the awarded marks	Response B, correct B, positive 2, negative `0.5`	The question is awarded exactly `+2.0`	As expected. Automated: `test_UT_053_correct_response_earns_full_positive_marks` (evaluation.tests): ok	**Pass**
UT_054	Evaluation	FR-023 — Deduct the configured negative marks for	Scoring engine callable; negative marking enabled at 0.5	1: Invoke the evaluator with a response of option A	Response A, correct B, positive 2, negative `0.5`	The question is awarded exactly `−0.5`	As expected. Automated: `test_UT_054_incorrect_response_loses`	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		an incorrect response		2: Read the awarded marks			`_the_configured_negative_marks` (evaluation.tests): ok	
UT_055	Evaluation	FR-023 — Award zero and apply no deduction to an unattempted question	Scoring engine callable; negative marking enabled at 0.5	1: Invoke the evaluator with no response recorded 2: Read the awarded marks	Response `null`, correct B, negative `0.5`	The question is awarded exactly `0.0`; the penalty is **not** applied — the boundary that separates skipping from guessing	As expected. Automated: `test_UT_055_unanswered_question_earns_zero_with_no_deduction` (evaluation.tests): ok	**Pass**
UT_056	Evaluation	FR-023 — Apply no deduction when negative marking is disabled	Scoring engine callable; negative marking disabled on the examination	1: Invoke the evaluator with an incorrect response 2: Read the awarded marks	Response A, correct B, negative marking off	The question is awarded exactly `0.0`, not a negative value	As expected. Automated: `test_UT_056_no_deduction_when_negative_marking_is_disabled` (evaluation.tests): ok	**Pass**
UT_057	Evaluation	FR-024 — Compute total, percentage and a Pass outcome at the pass mark	Scoring engine callable; pass mark 40%	1: Evaluate an attempt scoring 20 of 50 2: Read total, percentage and outcome	Total `20`, declared total `50`, pass mark `40%`	Total `20`, percentage `40.0%`, outcome **Pass** — the boundary is inclusive	As expected. Automated: `test_UT_057_score_exactly_at_the_pass_mark_is_a_pass` (evaluation.tests): ok	**Pass**
UT_058	Evaluation	FR-024 — Compute a Fail outcome just below the pass mark	Scoring engine callable; pass mark 40%	1: Evaluate an attempt scoring 19.5 of 50 2: Read the outcome	Total `19.5`, declared total `50`, pass mark `40%`	Percentage `39.0%`, outcome **Fail**	As expected. Automated: `test_UT_058_score_just_below_the_pass_mark_is_a_fail` (evaluation.tests): ok	**Pass**
IT_029	Evaluation	FR-022 — Every submitted attempt is evaluated without human intervention	`EXM_1` with three submitted attempts	1: Submit three attempts 2: Take no Faculty action of any kind 3: Query the evaluation records	Three submitted attempts	All three carry a computed score, percentage and outcome; none awaits a manual trigger or approval	As expected. Automated: `test_IT_029_every_submitted_attempt_is_evaluated_without_faculty_action` (evaluation.tests): ok	**Pass**
IT_030	Evaluation	FR-022, NFR-002 — A 100-question attempt is evaluated within 5 seconds	An examination of 100 questions with a submitted attempt; server timestamps available	1: Submit and record the server submission timestamp 2: Poll for the evaluation record and record its completion timestamp 3: Compute the elapsed time 4: Repeat over ten attempts	100-question paper, ten attempts	Every evaluation completes within 5 seconds of submission; the worst observed elapsed time is recorded	As expected. Automated: `test_IT_030_NFR_002_100_question_attempt_is_evaluated_within_5_seconds` (evaluation.test_performance): ok	**Pass**
IT_031	Evaluation	FR-023, FR-024 — A	An examination of 10	1: Submit an attempt	6 correct, 3	Total is `(6 × 2) − (3`	As expected. Automated:	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
		mixed attempt is scored correctly end to end	questions at +2 / −0.5 with a submitted attempt	with 6 correct, 3 incorrect and 1 unattempted 2: Query the computed total, percentage and outcome	incorrect, 1 skipped; +2 / −0.5; declared total 20; pass mark 40%	× 0.5) − 0 = 10.5; percentage 52.5%; outcome **Pass**; the skipped question contributes exactly zero	`test_IT_031_mixed_attempt_is_scored_end_to_end` (evaluation.tests): ok	
ST_011	Evaluation	FR-022, FR-023, FR-024 — End-to-end automatic scoring of a full cohort	TEST environment; EXM_1 with 25 questions; five enrolled Students	1: Have five Students sit and submit EXM_1 with differing answer patterns 2: Take no Faculty action 3: Inspect each attempt's score, percentage and outcome 4: Hand-compute two of the five from the answer key and compare	Five attempts across a 25-question paper	All five are evaluated automatically; the two hand-computed totals match the system's exactly; Pass/Fail is correct against the configured pass mark	As expected. Automated: `test_ST_011_five_students_are_scored_with_no_faculty_action` (evaluation.test_cohort): ok	**Pass**

UC-12 — Publish results and display the scorecard

Verifies FR-025, FR-026, FR-027.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_059	Results	FR-025 — An evaluated result is withheld until Faculty publishes	STU_A has a submitted, evaluated attempt on EXM_1; results not published	1: Sign in as STU_A 2: Open the dashboard and the examination 3: Look for a score	Evaluated but unpublished result	No score, percentage or outcome is shown; the examination reads "Results not yet published"	As expected. Automated: `test_UT_059_unpublished_result_is_withheld` (results.test_publish): ok	**Pass**
UT_060	Results	FR-026 — Publishing makes the result visible to the Student, and only once the window has closed	The situation of UT_059: results unpublished	1: Sign in as FAC_A 2: While the window of EXM_1 is still open, try to publish its results 3: After the window closes, publish the results 4: Sign in as STU_A and reopen the examination	EXM_1 results published	Step 2 is refused and the results stay unpublished; step 3 succeeds and the scorecard is now visible with total, percentage and outcome	As expected. Automated: `test_UT_060_publishing_makes_the_result_visible` (results.test_publish): ok	**Pass**
UT_061	Results	FR-026 — Unpublishing withdraws a previously visible result	EXM_1 results published and seen by STU_A	1: Sign in as FAC_A 2: Unpublish the results 3: Sign in as STU_A and reopen the examination	EXM_1 results unpublished	The scorecard is no longer shown; the examination reverts to "Results not yet published"; the stored	As expected. Automated: `test_UT_061_unpublishing_withdraws_it_and_keeps_the_score` (results.test_publish): ok	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
						score is unchanged		
UT_062	Results	FR-027 — The scorecard shows total, percentage, outcome and per-question marks	EXM_1 results published; STU_A scored 32 of 50	1: Sign in as STU_A 2: Open the scorecard 3: Inspect every displayed field	Total 32, declared 50, pass mark 40%, 25 questions	Total 32, percentage 64.0%, outcome **Pass**, and a per-question line for all 25 questions showing the marks obtained on each	As expected. Automated: test_UT_062_scorecard_shows_total_percentage_outcome_and_every_question (results.test_scorecard): ok	**Pass**
UT_063	Results	FR-027 — Per-question marks on the scorecard sum to the reported total	EXM_1 results published for STU_A	1: Open the scorecard 2: Sum the per-question marks 3: Compare with the displayed total	25 per-question values including negatives	The sum equals the displayed total exactly	As expected. Automated: test_UT_063_per_question_marks_sum_to_the_total (results.test_scorecard): ok	**Pass**
IT_032	Results	FR-025 — An unpublished result is unreachable through the API, not merely hidden in the UI	STU_A has an evaluated but unpublished result	1: Sign in as STU_A 2: Call GET /api/results/{attemptId} directly 3: Inspect the response body	STU_A's attempt ID; results unpublished	Refused with HTTP 403; no score, percentage or per-question mark appears anywhere in the body	As expected. Automated: test_IT_032_unpublished_result_is_403_through_the_api (results.test_publish): ok	**Pass**
IT_033	Results	FR-025 — One Student cannot read another's result	EXM_1 results published; STU_A and STU_B both attempted	1: Sign in as STU_A 2: Call the result endpoint with STU_B's attempt ID	STU_B's attempt ID	Refused with HTTP 403; STU_B's score is not disclosed	As expected. Automated: test_IT_033_another_students_result_is_403 (results.test_publish): ok	**Pass**
IT_034	Results	FR-026, FR-027 — The published scorecard matches the stored evaluation exactly	EXM_1 sat by five Students and published	1: Publish results 2: For each Student, open the scorecard 3: Compare each displayed total, percentage and outcome against the stored evaluation record	Five evaluated attempts	Every displayed figure matches its stored evaluation exactly; no rounding or transcription discrepancy	As expected. Automated: test_IT_034_every_published_scorecard_matches_the_stored_evaluation (results.test_scorecard): ok	**Pass**
ST_012	Results	FR-025, FR-026, FR-027 — End-to-end publication and scorecard	TEST environment; EXM_1 sat by five Students	1: Confirm no Student can see a score before publication 2: As FAC_A, publish 3: As each Student, open the scorecard and verify the figures 4: Unpublish and confirm the scorecards disappear	Five evaluated attempts	Results stay hidden until published, every scorecard matches the stored evaluation, and unpublishing withdraws visibility without altering the data	As expected. Automated: test_ST_012_end_to_end_publication_and_scorecard (results.test_scorecard): ok	**Pass**

UC-13 — Log integrity events and maintain the audit trail

Verifies FR-028, FR-029, FR-030.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_064	Integrity	FR-028 — A single focus-loss event is recorded with a timestamp	STU_A has an attempt in progress on EXM_1	1: Switch to another browser tab once 2: Return to the attempt 3: Query the integrity event log for the attempt	One tab switch	Exactly one focus-loss event is recorded against the attempt, carrying a server-side timestamp	As expected. Automated: `test_UT_064_one_focus_loss_is_one_event_with_a_server_time` (integrity.test_focus): ok	**Pass**
UT_065	Integrity	FR-028 — Repeated focus losses are each recorded separately	STU_A has an attempt in progress	1: Switch away and back three times 2: Query the event log	Three tab switches	Exactly three events are recorded, each with its own distinct timestamp; none is coalesced or dropped	As expected. Automated: `test_UT_065_three_losses_are_three_events` (integrity.test_focus): ok	**Pass**
UT_066	Integrity	FR-028 — No event is recorded when focus is never lost	STU_A has an attempt in progress	1: Complete the attempt without leaving the window 2: Query the event log	Zero tab switches	The event log for the attempt is empty; the detector does not fire on ordinary in-page interaction	As expected. Automated: `test_UT_066_no_loss_means_no_event` (integrity.test_focus): ok	**Pass**
UT_067	Integrity	FR-028 — The recorded timestamp is server-side, not client-supplied	STU_A has an attempt in progress	1: Set the client clock forward by one hour 2: Switch tabs once 3: Query the recorded event timestamp	Client clock +1 h; one tab switch	The stored timestamp reflects true server time, not the tampered client clock	As expected. Automated: `test_UT_067_client_supplied_time_is_ignored` (integrity.test_focus): ok	**Pass**
UT_068	Audit	FR-030 — An audit entry records actor, action, entity, timestamp and changed values	FAC_A signed in; audit log inspectable	1: Edit a question's stem and its answer key 2: Query the audit log for that question	Question Q_SC1, stem and correct option changed	One entry exists carrying the actor FAC_A, the action, the affected entity, a server timestamp, and both the previous and the new stem and answer key	As expected. Automated: `test_UT_068_an_edit_records_old_and_new_values` (repository.test_audit): ok `test_UT_068_question_edit_records_old_and_new_stem_and_answer_key` (repository.test_audit_coverage): ok	**Pass**
IT_035	Integrity	FR-029 — No correct-answer data reaches the candidate during an active attempt	STU_A has an attempt in progress on EXM_1	1: Capture every HTTP response delivered to the browser during the attempt 2: Search the payloads for `isCorrect`, `correctOption`,	EXM_1; known answer key	No response body, HTML attribute or client-side variable exposes which option is correct; the key is absent from everything the	As expected. Automated: `test_IT_035_paper_payload_never_contains_correct_answer_data` (attempts.test_timer_payload): ok	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				answerKey and the known correct values 3: Inspect the rendered HTML and client-side JavaScript state		browser receives		
IT_036	Audit	FR-030 — Every mutation class writes an audit entry	ADM_A and FAC_A signed in	1: Create a user, a course, a question and an examination, and submit an attempt 2: Query the audit log for each 3: Check who is recorded for that attempt's submission and evaluation 4: Change several examination fields at once 5: Publish results	One mutation of each of the five entity classes: User, Course, Question, Exam, Attempt	An entry exists for every one of the five, so no mutation class is silently unlogged; the submission is attributed to the Student and the evaluation to the system; every changed examination field is recorded; the results publication is recorded	As expected. Automated: `test_IT_036_every_mutation_class_writes_an_entry` (repository.test_audit): ok	**Pass**
IT_037	Audit	FR-030 — The audit log is append-only and resists modification	Audit entries exist; database reachable	1: Attempt to update an existing audit row through the application 2: Attempt to delete one 3: Inspect the table's constraints	An existing audit entry	Both attempts are refused; the application exposes no update or delete path for audit rows; the entry is unchanged	As expected. Automated: `test_IT_037_audit_rows_cannot_be_changed_or_deleted` (repository.test_audit): ok	**Pass**
ST_013	Integrity & Audit	FR-028, FR-029, FR-030 — End-to-end evidentiary trail across an examination cycle	TEST environment; EXM_1 live; STU_A enrolled	1: Sit EXM_1 as STU_A, switching tabs twice 2: Confirm no answer key appears in any captured payload 3: Submit 4: As FAC_A, publish results 5: As ADM_A, open the audit log filtered to EXM_1 and reconcile it against everything that occurred	One attempt, two focus-loss events, one publication	The log accounts for every state change — attempt creation, submission, evaluation, publication — the two integrity events are attached to the attempt for Faculty review, and no answer key ever reached the candidate	Automated: `test_ST_013_admin_reads_the_log_filtered_to_one_exam` (repository.test_audit): ok. Browser re-run (System Validation Report, 3 Nov): the audit log accounts for creation, submission, evaluation and publication, and no answer key reached the browser, but the attempt carries **6 focus-loss events for 2 tab switches**: every question page left adds one (Defect 1). After the fix (11 Nov, D-	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
							15), browser run 3: **2 focus-loss events for 2 tab switches**, still 2 after answering and submitting; audit log and payloads as before	